From Exact Dyadic Formulae to the Small-Argument Asymptotic of the Fabius Function

q-binomial decompression, endpoint–Laplace transfer, and the lower-Lambert phase

Technical companion to the formal development

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Abstract

The exact arithmetic of the Fabius function and its endpoint asymptotics are usually presented as two nearly disjoint subjects. At reciprocal powers of two one has rational recurrences, composition sums, Thue–Morse power sums, and a finite half-base q-binomial formula; near the origin one has a negative-Laplace product, a lower-Lambert saddle, and a logarithmically periodic correction. This report supplies the missing bridge.

The central exact observation is that the apparently formidable q-binomial numerator at $F(2^{-n})$ is not a new asymptotic object. After choosing a convenient scalar translation, the inner Thue–Morse block becomes a coefficient of the ratio $A(-2^k X)/A(-X)$, where

$$A(z) = \sum_{n\geq 0} a_n z^n = \prod_{j\geq 1} \frac{\mathrm{e}^{z/2^j} - 1}{z/2^j}, \qquad F(2^{-n}) = 2^{-\binom{n}{2}} a_n.$$

The Gaussian row at $q = 1/2$ is exactly the divided-difference functional on the geometric nodes $-1, -2, \ldots, -2^n$; it annihilates all lower powers and extracts the leading coefficient. Consequently the complete finite numerator collapses identically to $\mathfrak{m}_n a_n$, where $\mathfrak{m}_n = \prod_{j=1}^{n} (2^j - 1)$. This "decompression" proves that the q-Pochhammer, q-binomial, Thue–Morse, recurrence, composition, and generating-product formulae all encode the same coefficient sequence.

Two asymptotic derivations are then developed. The fully rigorous real-variable route writes the half moment as $d_n = \mathbb{E}(1 - X)^n$, compares it through second order with the negative Laplace transform $A(-n) = \mathbb{E}\mathrm{e}^{-nX}$, and uses the exact quadratic-plus-periodic decomposition of $\log A(-s)$. A complementary coefficient-saddle route applies Cauchy's formula directly to $a_n = [z^n]A(z)$; it reaches the same lower-Lambert coordinate and provides an independent explanation of every cancellation. If $L = \log 2$ and λ_n is the large solution of

$$\lambda_n - \frac{\log \lambda_n}{L} = n, \qquad \lambda_n = -\frac{1}{L} W_{-1}(-L 2^{-n}),$$

then the resulting sharp dyadic formula is

$$\log F(2^{-n}) = -\frac{L}{2}\lambda_n^2 + \left(1 + \frac{L}{2}\right)\lambda_n - \frac{1}{2}\log \lambda_n + C_F + \Psi(\lambda_n) + \mathcal{O}(n^{-1}),$$

where Ψ is a smooth, nonconstant, mean-zero, one-periodic function with explicit Gamma–zeta Fourier coefficients. Expanding the Lambert phase yields

$$\log F(2^{-n}) = -\frac{L}{2}n^2 - n\log n + \left(1 + \frac{L}{2}\right)n - \frac{\log^2 n}{2L} + C_F - \frac{\log^2 n}{2L^2 n} + \Psi(\lambda_n) + \mathcal{O}(n^{-1}).$$

The exact phase is essential: replacing it by $\log_2 n$ leaves a generally larger $(\log n)/n$ term. Corollaries give equally sharp asymptotics for the recurrence sequence, the half moments, the composition sum, and the literal q-binomial numerator. Numerical checks against exact rational values confirm the constant, phase, and first saddle correction.

Contents

1 The missing bridge

The Fabius function admits two descriptions that look, at first sight, almost antagonistic. Its values at dyadic rationals are algebraic and finite: every value is rational; there are terminating recurrences; the Thue–Morse sequence provides finite signed power sums; and a half-base Gaussian-binomial row packages those sums into a compact formula. Its endpoint asymptotic is analytic and infinite: one studies a Laplace product, separates a periodic renormalization of a dyadic harmonic sum, and evaluates a saddle determined by the lower real branch of Lambert's W-function.

The two subjects are not merely compatible. They are two coordinate systems on the same object. The exact formulae contain the asymptotic information in a highly compressed form, and the purpose of this report is to make the decompression explicit.

1.1 What is proved

There are four levels of connection.

1. The reciprocal-power value $F(2^{-n})$ is exactly a half moment divided by a factorial and a triangular power of two. This already gives the quadratic logarithmic decay.
2. After factorial normalization, all reciprocal-power values are coefficients of one entire function A. The recurrence, composition sum, and infinite product are equivalent presentations of A.
3. The finite q-binomial/Thue–Morse formula can be reduced identically to the same coefficient $a_n = [z^n]A(z)$. No limiting argument is involved. The q-binomial row is a geometric divided difference; its role is to undo the blockwise Thue–Morse cancellation.
4. The product for $A(-s)$ has an exact quadratic-plus-periodic logarithm. Transferring from $A(-n)$ to the endpoint moment, or applying a coefficient saddle directly to A, yields the corrected lower-Lambert asymptotic.

The first, second, third, and the full real-variable fourth level are represented in the Lean formalization under `Analysis/FabiusFunction`. The coefficient-saddle calculation in Section 8 is included as an independent conventional-mathematics derivation; it is useful because it starts from the coefficient formula itself and exposes the same cancellations with very little probability language.

1.2 Headline notation and result

Throughout,

$$L = \log 2. \tag{1}$$

Let P be the periodic correction in the exact negative-Laplace decomposition, let

$$\overline{P} = \int_0^1 P(u)\,\mathrm{d}u, \qquad \Psi(u) = P(u) - \overline{P}, \tag{2}$$

and put

$$A_0 = \frac{6\gamma^2 + 12\gamma_1 - \pi^2}{12}, \qquad C_F = \frac{A_0}{L} - \frac{7L}{12} - \frac{1}{2}\log\pi. \tag{3}$$

Here γ is Euler's constant and γ_1 is the first Stieltjes constant. For every sufficiently large integer n, let λ_n be the large positive solution of

$$\lambda_n 2^{-\lambda_n} = 2^{-n}, \qquad \text{equivalently} \qquad \lambda_n - \frac{\log\lambda_n}{L} = n. \tag{4}$$

Theorem 1.1 (Sharp reciprocal-power asymptotic). *As $n \to \infty$,*

$$\boxed{\log F(2^{-n}) = -\frac{L}{2}\lambda_n^2 + \left(1 + \frac{L}{2}\right)\lambda_n - \frac{1}{2}\log\lambda_n + C_F + \Psi(\lambda_n) + \mathcal{O}(n^{-1}).} \tag{5}$$

Equivalently,

$$\begin{aligned} \log F(2^{-n}) = {} & -\frac{L}{2}n^2 - n\log n + \left(1 + \frac{L}{2}\right)n - \frac{\log^2 n}{2L} + C_F \\ & - \frac{\log^2 n}{2L^2 n} + \Psi(\lambda_n) + \mathcal{O}(n^{-1}). \end{aligned} \tag{6}$$

The function Ψ is smooth, mean-zero, one-periodic, and nonconstant. Its Fourier coefficients are explicit.

The rest of the report derives Theorem 1.1 from the exact arithmetic formulae rather than taking it as an externally supplied endpoint theorem.

The most important conceptual point is that the periodic term is already present in the exact dyadic values. It does not arise because one interpolates between dyadic points. The fractional parts of the exact saddle coordinates λ_n drift through the logarithmic period, so the single rational sequence $F(2^{-n})$ samples the entire periodic correction.

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2 One coefficient sequence behind the exact formulae

2.1 The half moments and the endpoint identity

Let F denote the bounded Fabius distribution function and let up be Rvachev's even compactly supported density in the normalization

$$\mathrm{up}(t) = F(1 - |t|), \qquad |t| \le 1. \tag{7}$$

Define the half moments

$$d_0 = 1, \qquad d_n = n\int_0^1 t^{n-1}\,\mathrm{up}(t)\,\mathrm{d}t \quad (n \ge 1). \tag{8}$$

They are rational and positive. Their first few values are

$$d_0 = 1, \qquad d_1 = \frac{1}{2}, \qquad d_2 = \frac{5}{18}, \qquad d_3 = \frac{1}{6}, \qquad d_4 = \frac{143}{1350}. \tag{9}$$

Theorem 2.1 (Exact endpoint–moment bridge). *For every $n \ge 0$,*

$$d_n = n!\,2^{\binom{n}{2}}F(2^{-n}), \qquad F(2^{-n}) = \frac{d_n}{n!\,2^{\binom{n}{2}}}. \tag{10}$$

Proof. The primitive-chain proof is especially revealing. For $r \ge 0$ define

$$f_r(t) = 2^{\binom{r}{2}}\,\mathrm{up}\big((t+1)2^{-r} - 1\big), \qquad -1 \le t \le 1. \tag{11}$$

The functional–differential equation of up implies

$$f'_{r+1}(t) = f_r(t), \qquad f_r(-1) = 0. \tag{12}$$

Put $J_n = \int_{-1}^{0} t^{n-1}\,\mathrm{up}(t)\,\mathrm{d}t$. Repeated integration by parts, using (12), gives

$$J_n = (-1)^{n-1}(n-1)!\,f_n(0). \tag{13}$$

Evenness of up turns the left side into $(-1)^{n-1}\int_0^1 t^{n-1}\,\mathrm{up}(t)\,\mathrm{d}t$, while

$$f_n(0) = 2^{\binom{n}{2}}\,\mathrm{up}(2^{-n}-1) = 2^{\binom{n}{2}}F(2^{-n}).$$

Multiplication by n proves (10). The case $n=0$ is $F(1) = d_0 = 1$. □

There is also a probability interpretation that will be decisive later. Let X be a random variable with distribution function F. Integrating by parts against the distribution measure gives

$$d_n = \mathbb{E}(1-X)^n. \tag{14}$$

Thus d_n is simultaneously an exact rational endpoint value and an endpoint moment of the probability law.

2.2 Factorial normalization and the recurrence

Set

$$a_n = \frac{d_n}{n!}, \qquad A(z) = \sum_{n=0}^{\infty} a_n z^n. \tag{15}$$

Then Theorem 2.1 becomes the particularly economical identity

$$\boxed{F(2^{-n}) = 2^{-\binom{n}{2}}a_n.} \tag{16}$$

The half-moment recurrence is equivalent to

$$(2^n-1)a_n = \sum_{k=0}^{n-1}\frac{a_k}{(n-k+1)!}, \qquad n \ge 1, \qquad a_0 = 1. \tag{17}$$

Equivalently, directly in terms of endpoint values,

$$F(2^{-n}) = \frac{2^{-\binom{n}{2}}}{2^n-1}\sum_{k=0}^{n-1}\frac{2^{\binom{k}{2}}}{(n-k+1)!}F(2^{-k}). \tag{18}$$

The recurrence is not merely an evaluation algorithm. It is already a coefficient equation. Let

$$E(z) = \frac{\mathrm{e}^z-1}{z}, \qquad E(0) = 1. \tag{19}$$

Multiplying (17) by z^n and summing gives

$$\boxed{A(2z) = E(z)A(z).} \tag{20}$$

Conversely, coefficient comparison in (20) recovers (17). Thus the exact recurrence and the analytic refinement equation are the same statement.

2.3 Composition sum

Repeated substitution in (17) gives a finite closed form. A composition $\rho \vDash n$ is an ordered tuple $\rho = (r_1, \dots, r_m)$ of positive integers with sum n; write $s_j = r_1 + \cdots + r_j$ for its prefix sums.

Proposition 2.2 (Exact composition formula)**.** *For $n \geq 1$,*

$$a_n = \sum_{\rho \vDash n} \prod_{j=1}^{\ell(\rho)} \frac{1}{(2^{s_j} - 1)(r_j + 1)!}. \tag{21}$$

Consequently

$$F(2^{-n}) = 2^{-\binom{n}{2}} \sum_{\rho \vDash n} \prod_{j=1}^{\ell(\rho)} \frac{1}{(2^{s_j} - 1)(r_j + 1)!}. \tag{22}$$

Proof. Every use of (17) replaces a current index s_j by a smaller prefix s_{j-1} and contributes $((2^{s_j} - 1)(s_j - s_{j-1} + 1)!)^{-1}$. A complete descent from n to 0 is therefore the same as a composition of n, with increments $r_j = s_j - s_{j-1}$. Summing over all descents gives (21). □

This formula looks combinatorial, but asymptotically it is not a sum over independently comparable contributions. The number of compositions is 2^{n-1}, and the dominant family moves with n. The correct way to analyze the sum is to reassemble it into A.

2.4 The entire product

Iterating (20) toward the origin gives

$$\boxed{A(z) = \prod_{j=1}^{\infty} E(z/2^j) = \prod_{j=1}^{\infty} \frac{e^{z/2^j} - 1}{z/2^j}.} \tag{23}$$

The product converges locally uniformly and defines an entire function. It is the moment generating function of X:

$$A(z) = \mathbb{E}e^{zX}. \tag{24}$$

Since $E(z) = e^z E(-z)$ and $\sum_{j \geq 1} 2^{-j} = 1$, the product also satisfies the exact reflection identity

$$A(z) = e^z A(-z). \tag{25}$$

The equivalence graph is now

$$\boxed{\begin{array}{c}\text{endpoint values} \longleftrightarrow d_n \longleftrightarrow a_n \longleftrightarrow \text{recurrence} \\ \longleftrightarrow \text{composition sum} \longleftrightarrow A(z) \longleftrightarrow \text{entire product}\end{array}}$$

The next section inserts the q-binomial and Thue–Morse formula into this same graph.

3 Decompressing the half-base q-binomial formula

The q-binomial formula is the most conspicuous exact representation and the least transparent asymptotic one. Its finite sum contains two distinct annihilation mechanisms:

- inside each dyadic block, Thue–Morse signs annihilate low polynomial degrees;
- across the block sizes, a Gaussian-binomial divided difference annihilates the remaining low powers of the geometric scale.

Once both cancellations are written as coefficient extraction, the entire numerator reduces to $\mathfrak{m}_n a_n$.

3.1 The geometric interpolation functional

For $q = 1/2$, write

$$(a;1/2)_n = \prod_{j=0}^{n-1}(1 - a2^{-j}), \tag{26}$$

and let $\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}$ denote the Gaussian binomial coefficient [3, 9, 12, 6]. The finite q-binomial theorem is

$$(z;1/2)_n = \sum_{k=0}^{n}(-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} z^k. \tag{27}$$

Define

$$x_k = -2^k, \qquad w_{n,k} = (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2}, \qquad \mathfrak{m}_n = \prod_{j=1}^{n}(2^j - 1). \tag{28}$$

Lemma 3.1 (Leading-coefficient extraction). *If R is a polynomial of degree at most n, then*

$$\sum_{k=0}^{n} w_{n,k} R(-2^k) = \mathfrak{m}_n [z^n] R(z). \tag{29}$$

Equivalently,

$$\sum_{k=0}^{n} w_{n,k} x_k^d = \begin{cases} 0, & 0 \le d < n, \\ \mathfrak{m}_n, & d = n. \end{cases} \tag{30}$$

Proof. The normalized weights are the barycentric weights for the nodes $x_0, \ldots, x_n$:

$$\frac{w_{n,k}}{\mathfrak{m}_n} = \frac{1}{\prod_{j \neq k}(x_k - x_j)}. \tag{31}$$

The coefficient of z^n in the Lagrange interpolation formula for R is therefore the left side of (29) divided by $\mathfrak{m}_n$. Formula (31) can also be checked directly from (27). □

The Pochhammer normalization is exactly the same Mersenne product:

$$(1/2;1/2)_n = 2^{-\binom{n+1}{2}} \mathfrak{m}_n. \tag{32}$$

This identity will make the last power-of-two cancellation automatic.

3.2 The finite Thue–Morse blocks

Let

$$\varepsilon_r = (-1)^{\mathrm{wt}_2(r)}. \tag{33}$$

For $k, N \ge 0$ and a scalar translation Q, define

$$S_{k,N}(Q) = \sum_{r=0}^{2^k-1} \varepsilon_r (r - 2^k + Q)^N. \tag{34}$$

The complete-block exponential generating function is

$$C_k(X) = \sum_{r<2^k} \varepsilon_r \mathrm{e}^{(r-2^k)X}. \tag{35}$$

The elementary binary product

$$\sum_{r<2^k} \varepsilon_r z^r = \prod_{j=0}^{k-1}(1 - z^{2^j}) \tag{36}$$

implies

$$\frac{C_k(X)}{X^k} = (-1)^k 2^{\binom{k}{2}} \mathrm{e}^{-X} \prod_{j=0}^{k-1} E(-2^j X). \tag{37}$$

In particular, C_k has a zero of order k at the origin. Equivalently, $S_{k,N}(Q) = 0$ for $N < k$, independently of Q.

Iterating (20) on the negative dyadic orbit gives

$$\frac{A(-2^k X)}{A(-X)} = \prod_{j=0}^{k-1} E(-2^j X). \tag{38}$$

Translation invariance permits the especially convenient choice $Q = 1$. Since

$$\sum_{N\geq 0} S_{k,N}(1)\frac{X^N}{N!} = \mathrm{e}^X C_k(X), \tag{39}$$

(37) and (38) yield the exact coefficient identity

$$\boxed{\frac{S_{k,n+k}(1)}{(n+k)!} = (-1)^k 2^{\binom{k}{2}} [X^n]\frac{A(-2^k X)}{A(-X)}.} \tag{40}$$

This is the first decompression: an inner Thue–Morse power sum is a coefficient of the same entire function that generates the half moments.

3.3 The literal numerator collapses

At $m = 1$, the global q-binomial formula reads

$$F(2^{-n}) = \frac{\mathcal{N}_n}{2^{n^2}(1/2; 1/2)_n}, \tag{41}$$

where, using the shift $Q = 1$,

$$\mathcal{N}_n = \sum_{k=0}^{n} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!} S_{k,n+k}(1). \tag{42}$$

Theorem 3.2 (q-binomial decompression)**.** *For every* $n \geq 0$,

$$\boxed{\mathcal{N}_n = \mathfrak{m}_n a_n.} \tag{43}$$

Consequently the finite q-binomial formula is identically the endpoint formula $F(2^{-n}) = 2^{-\binom{n}{2}} a_n$.

Proof. For fixed n, introduce the polynomial

$$R_n(z) = [X^n]\frac{A(zX)}{A(-X)}. \tag{44}$$

It has degree at most n. Since $A(-X)^{-1} = 1 + \mathcal{O}(X)$ and $A(zX) = \sum_{j\geq 0} a_j z^j X^j$, its leading coefficient is

$$[z^n]R_n(z) = a_n. \tag{45}$$

Substitution of (40) into (42) gives

$$\begin{aligned} \mathcal{N}_n &= \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} 2^{-2\binom{k}{2}} (-1)^k 2^{\binom{k}{2}} R_n(-2^k) \\ &= \sum_{k=0}^{n} w_{n,k} R_n(-2^k). \end{aligned} \tag{46}$$

By Theorem 3.1, this is $\mathfrak{m}_n [z^n] R_n(z) = \mathfrak{m}_n a_n$, proving (43). Finally, (32) gives

$$\frac{\mathfrak{m}_n a_n}{2^{n^2} (1/2; 1/2)_n} = \frac{\mathfrak{m}_n a_n}{2^{n^2 - \binom{n+1}{2}} \mathfrak{m}_n} = 2^{-\binom{n}{2}} a_n.$$

□

The two finite cancellations have now been separated completely. The Thue–Morse block removes k powers of X and converts a power sum into $R_n(-2^k)$. The half-base q-binomial row then removes all powers of the scale node below degree n and returns the coefficient a_n. The q-Pochhammer factor does not introduce a new special-function asymptotic; it is the normalization of this divided difference.

3.4 General dyadic numerators

For completeness, the exact global formula is

$$\begin{aligned} \mathcal{F}(m/2^n) = \frac{1}{2^{n^2} (1/2; 1/2)_n} \sum_{k=0}^{n} \frac{\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \\ \times \sum_{r=0}^{m2^k - 1} \varepsilon_r (r - m2^k + Q)^{n+k}. \end{aligned} \tag{47}$$

It is independent of Q. Splitting $r = h2^k + s$ factors the long block into the complete block C_k and a finite prefix polynomial. The same interpolation functional then extracts one coefficient of a product of that prefix polynomial with A. Thus even the general formula contains no asymptotically unrelated object. What changes when m varies with n is the coefficient multiplier, not the underlying dyadic product.

4 The first asymptotic already visible in the exact values

Before resolving the periodic fluctuation, the endpoint identity alone gives the principal quadratic law. Taking logarithms in (10),

$$\log F(2^{-n}) = \log d_n - \log(n!) - \binom{n}{2} L. \tag{48}$$

Because $0 \le \mathrm{up} \le 1$,

$$d_n \le n \int_0^1 t^{n-1} \, \mathrm{d}t = 1. \tag{49}$$

A fixed positive portion of the mass on $[0, 1/2]$ gives, for $n \ge 1$,

$$2^{-(n+1)} \le d_n. \tag{50}$$

Combining these bounds with elementary factorial estimates yields the explicit inequality

$$\left|\log F(2^{-n}) + \frac{L}{2}n^2\right| \le 3n\log(n+1), \qquad n \ge 1. \tag{51}$$

Therefore

$$\lim_{n\to\infty} \frac{\log F(2^{-n})}{n^2} = -\frac{L}{2}. \tag{52}$$

This is already a derivation of the leading endpoint asymptotic from exact arithmetic. It also explains why neither the factorial nor the half moment can change the coefficient of n^2: $\log(n!) = \mathcal{O}(n\log n)$ and $\log d_n = \mathcal{O}(n)$ under the crude bounds. To recover the $-n\log n$ term, the $\log^2 n$ term, the constant, and the periodic fluctuation, however, one must understand d_n much more sharply.

5 The exact negative-Laplace decomposition

The sharp asymptotic enters through the same function A. For $s > 0$ put

$$B(s) = A(-s) = \mathbb{E}\mathrm{e}^{-sX}, \qquad \Lambda(s) = \log B(s). \tag{53}$$

From (23),

$$B(s) = \prod_{j=1}^{\infty} \frac{1 - \mathrm{e}^{-s/2^j}}{s/2^j}, \tag{54}$$

and hence

$$\Lambda(s) = \sum_{j=1}^{\infty} \kappa(s/2^j), \qquad \kappa(u) = \log\frac{1-\mathrm{e}^{-u}}{u}. \tag{55}$$

This is a dyadic harmonic sum. Its dilation equation is exact:

$$\Lambda(2s) = \Lambda(s) + \log(1 - \mathrm{e}^{-s}) - \log s. \tag{56}$$

5.1 Periodization of the dilation equation

Define the forward logarithmic tail

$$T(s) = \sum_{m=0}^{\infty} \log(1 - \mathrm{e}^{-s2^m}) < 0, \qquad \mathcal{E}(s) = -T(s) > 0. \tag{57}$$

Since $2^m \ge m+1$,

$$0 \le \mathcal{E}(s) \le \frac{\mathrm{e}^{-s}}{(1-\mathrm{e}^{-s})^2}, \tag{58}$$

and in particular $\mathcal{E}(s) \le 4\mathrm{e}^{-s}$ for $s \ge L$. For $u \in \mathbb{R}$, set

$$P(u) = \Lambda(2^u) + \frac{L}{2}(u^2 - u) + T(2^u). \tag{59}$$

Theorem 5.1 (Exact quadratic-plus-periodic decomposition). *The function P is smooth and one-periodic. For every $s > 0$,*

$$\boxed{\Lambda(s) = -\frac{(\log s)^2}{2L} + \frac{1}{2}\log s + P(\log_2 s) + \mathcal{E}(s).} \tag{60}$$

All derivatives of P are bounded.

Proof. The forward tail satisfies

$$T(2s) = T(s) - \log(1 - \mathrm{e}^{-s}).$$

Under $u \mapsto u + 1$, the change in the quadratic term of (59) is

$$\frac{L}{2}\big((u+1)^2 - (u+1) - (u^2 - u)\big) = Lu = \log s.$$

This cancels the changes in (56) and in T, proving $P(u+1) = P(u)$. Rearrangement gives (60). Local uniform convergence of the defining series permits differentiation; periodicity then makes every derivative bounded. □

The drift $-(\log s)^2/(2L)$ is the analytic counterpart of the triangular power $2^{-\binom{n}{2}}$ in the exact endpoint formula. The periodic remainder is the residual freedom in solving the dilation equation on logarithmic scale.

5.2 Mean, Fourier modes, and the asymptotic constant

Let $\overline{P}$ and Ψ be as in (2). The Mellin transform of the kernel gives

$$\overline{P} = \frac{A_0}{L} - \frac{L}{12}, \tag{61}$$

where A_0 is defined in (3). With

$$\chi_k = \frac{2\pi \mathrm{i} k}{L}, \tag{62}$$

the Fourier coefficients of the centered function are

$$\widehat{\Psi}(0) = 0, \qquad \boxed{\widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{L}} \qquad (k \neq 0). \tag{63}$$

The Gamma factor has no zeros, and ζ has no zeros on $\Re z = 1$ away from its pole at 1, so every nonzero Fourier coefficient is nonzero. Thus Ψ is genuinely nonconstant. Stirling's vertical estimate for Γ makes the coefficients exponentially small; in fact the oscillation has amplitude only a few parts in 10^6, but it cannot be deleted at an error scale tending to zero.

The Gaussian normalization later contributes $-\frac{1}{2}\log(2\pi)$. Therefore

$$\overline{P} - \frac{1}{2}\log(2\pi) = \frac{A_0}{L} - \frac{7L}{12} - \frac{1}{2}\log \pi = C_F. \tag{64}$$

Numerically,

$$\overline{P} = -1.10904536543418668\ldots, \qquad C_F = -2.02798389863885942\ldots. \tag{65}$$

6 Sharp route I: endpoint moments versus Laplace moments

This route begins with the exact rational object $d_n = \mathbb{E}(1-X)^n$ and converts it into the negative-Laplace product at the natural scale $s = n$. It is the route closest to the current Lean bridge modules.

6.1 Second-order kernel comparison

For $0 \le x \le 1$ and $n \ge 1$, the logarithmic expansion

$$n \log(1-x) = -nx - \frac{n}{2}x^2 + \mathcal{O}(nx^3)$$

is useful only after the large-x region is controlled uniformly. A convenient global form is

$$\left|(1-x)^n - \mathrm{e}^{-nx}\left(1 - \frac{n}{2}x^2\right)\right| \le 16\mathrm{e}^{-nx}(nx^3 + n^2x^4). \tag{66}$$

Integrating against the law of X gives

$$d_n = B(n) - \frac{n}{2}\mathbb{E}\big(X^2\mathrm{e}^{-nX}\big) + R_n, \tag{67}$$

$$|R_n| \le 16\left(n\mathbb{E}(X^3\mathrm{e}^{-nX}) + n^2\mathbb{E}(X^4\mathrm{e}^{-nX})\right). \tag{68}$$

Let $q(s) = \Lambda(s) = \log B(s)$. The normalized tilted second moment is

$$\frac{\mathbb{E}(X^2\mathrm{e}^{-sX})}{B(s)} = q''(s) + q'(s)^2. \tag{69}$$

The corresponding third- and fourth-moment identities are polynomials in q', q'', $q^{(3)}$, $q^{(4)}$. Differentiation of (60), together with the exponentially small tail, shows that these normalized moments have precisely the inverse powers needed to make (68) a relative $\mathcal{O}(n^{-1})$ error after the second term is retained.

Proposition 6.1 (Endpoint–Laplace logarithmic transfer)**.** *As $n \to \infty$,*

$$\boxed{\log d_n = q(n) - \frac{n}{2}\big(q''(n) + q'(n)^2\big) + \mathcal{O}(n^{-1}).} \tag{70}$$

Proof architecture. Divide (67) by $B(n)$. The second-order displacement is

$$\alpha_n = \frac{n}{2}\big(q''(n) + q'(n)^2\big) = \mathcal{O}(\log^2 n/n),$$

and the normalized remainder is $\mathcal{O}(n^{-1})$. Thus $d_n/B(n) = 1 - \alpha_n + \mathcal{O}(n^{-1})$. The local estimate $\log(1-y) = -y + \mathcal{O}(y^2)$ applies; since $\alpha_n^2 = \mathcal{O}(\log^4 n/n^2) = \mathcal{O}(n^{-1})$, it gives (70). □

6.2 Differentiating the exact periodic decomposition

Write

$$\ell = \log s, \qquad u = \log_2 s = \frac{\ell}{L}. \tag{71}$$

Ignoring an exponentially small differentiated tail, (60) gives

$$q'(s) = \frac{-\ell/L + 1/2 + \Psi'(u)/L}{s} + \mathcal{O}(\mathrm{e}^{-s}), \tag{72}$$

and

$$q''(s) = \frac{\ell/L - 1/2 - 1/L - \Psi'(u)/L + \Psi''(u)/L^2}{s^2} + \mathcal{O}(\mathrm{e}^{-s}). \tag{73}$$

Squaring (72) and adding (73), the terms linear in ℓ that do not involve Ψ' cancel. Since Ψ' and Ψ'' are bounded,

$$\frac{s}{2}\big(q''(s) + q'(s)^2\big) = \frac{\ell^2}{2L^2 s} - \frac{\ell\,\Psi'(u)}{L^2 s} + \mathcal{O}(s^{-1}). \tag{74}$$

This cancellation is the analytic reason that the first nonperiodic endpoint correction is $-\log^2 n/(2L^2 n)$ rather than a mixture of $\log^2 n/n$ and $\log n/n$ terms.

Substituting (60) and (74) into (70) yields the first sharp form.

Theorem 6.2 (Half moments at the logarithmic phase)**.** *Let* $u_n = \log_2 n$. *Then*

$$\begin{aligned}\log d_n = & -\frac{\log^2 n}{2L} + \frac{1}{2}\log n + \overline{P} + \Psi(u_n) \\ & -\frac{\log^2 n}{2L^2 n} + \frac{\log n}{L^2 n}\Psi'(u_n) + \mathcal{O}(n^{-1}).\end{aligned} \tag{75}$$

Combining (75) with Stirling's formula and (48) gives

$$\begin{aligned}\log F(2^{-n}) = & -\frac{L}{2}n^2 - n\log n + \left(1 + \frac{L}{2}\right)n - \frac{\log^2 n}{2L} + C_F \\ & -\frac{\log^2 n}{2L^2 n} + \Psi(u_n) + \frac{\log n}{L^2 n}\Psi'(u_n) + \mathcal{O}(n^{-1}).\end{aligned} \tag{76}$$

Formula (76) has already recovered the constant and the periodic function from the exact endpoint values. The derivative term is not an extra oscillation; it is the first-order displacement from the naive phase u_n to the exact Lambert phase.

7 The lower-Lambert phase and the derivative cancellation

7.1 The exact phase

The natural saddle coordinate at $x = 2^{-n}$ is

$$\lambda_n = -\frac{1}{L}W_{-1}(-L2^{-n}), \tag{77}$$

where W_{-1} is the lower real branch of Lambert's function [5]. It is the large solution of (4). Its fixed-point form is

$$\lambda_n = n + \log_2 \lambda_n. \tag{78}$$

The standard lower-Lambert expansion gives

$$\begin{aligned}\lambda_n = n + \frac{\log n}{L} + \frac{\log n}{L^2 n} + \frac{\log n - \frac{1}{2}\log^2 n}{L^3 n^2} \\ + \frac{\log n - \frac{3}{2}\log^2 n + \frac{1}{3}\log^3 n}{L^4 n^3} + \mathcal{O}\left(\frac{\log^4 n}{n^4}\right).\end{aligned} \tag{79}$$

For the periodic phase transfer only the first displacement is needed. Define

$$\delta_n = \log_2 \lambda_n - \log_2 n. \tag{80}$$

Then

$$\delta_n = \frac{\log n}{L^2 n} + \mathcal{O}\left(\frac{\log^2 n}{n^2}\right). \tag{81}$$

7.2 Why Ψ is evaluated at λ_n

Since n is an integer, periodicity and (78) imply the exact identity

$$\Psi(\lambda_n) = \Psi(\lambda_n - n) = \Psi(\log_2 \lambda_n). \tag{82}$$

Taylor expansion at $u_n = \log_2 n$, with uniformly bounded second derivative, gives

$$\begin{aligned}\Psi(\lambda_n) &= \Psi(u_n + \delta_n) \\ &= \Psi(u_n) + \frac{\log n}{L^2 n}\Psi'(u_n) + \mathcal{O}(n^{-1}).\end{aligned} \tag{83}$$

Indeed, the error in (81) contributes $\mathcal{O}(\log^2 n/n^2) = \mathcal{O}(n^{-1})$, and the quadratic Taylor remainder contributes the same or less. Substitution of (83) into (76) proves (6).

This calculation explains exactly why the phase may not be replaced by $\log_2 n$ at the claimed precision. One would omit the term

$$\frac{\log n}{L^2 n}\Psi'(\log_2 n), \tag{84}$$

which is generally of order $(\log n)/n$, not $1/n$. The tiny amplitude of Ψ affects numerical visibility, but not the asymptotic order.

Warning 7.1 (Phase precision is amplified). The leading exponent contains $-L\lambda^2/2$. An error $\Delta\lambda$ in the phase changes the exponent by approximately $-L\lambda\Delta\lambda$. Therefore deriving an elementary expansion with remainder $\mathcal{O}(n^{-1})$ requires the Lambert phase to one order beyond a naive term count. Keeping λ_n exact inside Ψ avoids this bookkeeping hazard.

7.3 Compact versus expanded form

Define

$$\mathcal{M}_n = -\frac{L}{2}\lambda_n^2 + \left(1+\frac{L}{2}\right)\lambda_n - \frac{1}{2}\log\lambda_n + C_F + \Psi(\lambda_n). \tag{85}$$

Substitution of (79) and collection of terms gives

$$\begin{aligned}\mathcal{M}_n = & -\frac{L}{2}n^2 - n\log n + \left(1+\frac{L}{2}\right)n - \frac{\log^2 n}{2L} + C_F \\ & -\frac{\log^2 n}{2L^2 n} + \Psi(\lambda_n) + \mathcal{O}(n^{-1}).\end{aligned} \tag{86}$$

Thus (5) and (6) are equivalent at the stated precision. The compact formula is structurally cleaner and is the one that survives unchanged for arbitrary small real arguments.

8 Sharp route II: a coefficient saddle from the exact generating product

The endpoint–Laplace route starts from $d_n = \mathbb{E}(1-X)^n$. There is a second route beginning directly with the recurrence coefficient $a_n = [z^n]A(z)$. It is particularly well suited to the question posed here because a_n is exactly what remains after the q-binomial formula is decompressed.

8.1 Positive-axis growth of A

The reflection identity (25) and Theorem 5.1 give, for $r > 0$,

$$\begin{aligned}K(r) := \log A(r) &= r + \Lambda(r) \\ &= r - \frac{\log^2 r}{2L} + \frac{1}{2}\log r + P(\log_2 r) + \mathcal{E}(r).\end{aligned} \tag{87}$$

The reference saddle is chosen by ignoring the bounded derivative of the periodic term:

$$r - \frac{\log r}{L} = n. \tag{88}$$

Its large solution is exactly $r = \lambda_n$. The exact coefficient saddle for $rK'(r) = n$ differs from λ_n by only $\mathcal{O}(1)$, because

$$rK'(r) = r - \frac{\log r}{L} + \frac{1}{2} + \frac{1}{L}P'(\log_2 r) + \mathcal{O}(re^{-r}). \tag{89}$$

An $\mathcal{O}(1)$ displacement of a saddle of curvature $\asymp 1/r$ changes the logarithmic coefficient estimate by only $\mathcal{O}(1/r)$.

8.2 Cauchy extraction

Cauchy's formula gives

$$a_n = \frac{A(r)}{2\pi r^n} \int_{-\pi}^{\pi} \frac{A(re^{i\theta})}{A(r)} e^{-in\theta}\, d\theta. \tag{90}$$

At $r = \lambda_n$, split the product (23) near the dyadic index $j \approx \log_2 r$. In a central window, Taylor expansion of the logarithm gives a Gaussian with variance parameter $r + \mathcal{O}(1)$; outside that window, a positive proportion of factors has modulus strictly below its positive-axis value, and the remaining factors supply arbitrary polynomial decay. The standard coefficient-saddle estimate [10, 8] is therefore

$$a_n = \frac{A(\lambda_n)}{\lambda_n^n \sqrt{2\pi\lambda_n}} \left(1 + \mathcal{O}(\lambda_n^{-1})\right). \tag{91}$$

The use of the reference rather than exact saddle is absorbed by the same relative error.

Taking logarithms and applying (87),

$$\log a_n = \lambda_n - n \log \lambda_n - \frac{\log^2 \lambda_n}{2L} + P(\log_2 \lambda_n) - \frac{1}{2}\log(2\pi) + \mathcal{O}(n^{-1}). \tag{92}$$

The $+\frac{1}{2}\log \lambda_n$ in K has canceled the Gaussian denominator. Now use $n = \lambda_n - (\log \lambda_n)/L$:

$$\log a_n = \lambda_n - \lambda_n \log \lambda_n + \frac{\log^2 \lambda_n}{2L} + P(\lambda_n) - \frac{1}{2}\log(2\pi) + \mathcal{O}(n^{-1}), \tag{93}$$

where periodicity was used in the form $P(\log_2 \lambda_n) = P(\lambda_n - n) = P(\lambda_n)$.

Finally, subtract $\binom{n}{2}L$ according to (16). Writing $\ell_n = \log \lambda_n$ and $n = \lambda_n - \ell_n/L$, one has

$$-\binom{n}{2}L = -\frac{L}{2}\lambda_n^2 + \lambda_n \ell_n - \frac{\ell_n^2}{2L} + \frac{L}{2}\lambda_n - \frac{1}{2}\ell_n.$$

Adding (93) makes the terms $\lambda_n \ell_n$ and $\ell_n^2/(2L)$ cancel exactly. Using (64) gives

$$\log F(2^{-n}) = -\frac{L}{2}\lambda_n^2 + \left(1 + \frac{L}{2}\right)\lambda_n - \frac{1}{2}\log \lambda_n + C_F + \Psi(\lambda_n) + \mathcal{O}(n^{-1}). \tag{94}$$

This is (5).

The coefficient saddle provides a particularly direct answer to the original question. Starting from the exact q-binomial value, Theorem 3.2 produces a_n; Cauchy extraction of that coefficient from the exact product produces the lower-Lambert asymptotic. The endpoint–Laplace route and the coefficient route are not the same proof in disguise: one compares $(1 - X)^n$ with e^{-nX} on the real line, while the other analyzes a complex coefficient integral. Their agreement fixes the normalization and phase very rigidly.

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9 Consequences for every exact arithmetic representation

9.1 The normalized recurrence and composition sum

Since $a_n = 2^{\binom{n}{2}} F(2^{-n})$, (6) immediately gives

$$\begin{aligned} \log a_n = &-n\log n + n - \frac{\log^2 n}{2L} + C_F \\ &- \frac{\log^2 n}{2L^2 n} + \Psi(\lambda_n) + \mathcal{O}(n^{-1}). \end{aligned} \tag{95}$$

Thus the exact composition sum (21) has the asymptotic

$$\sum_{\rho \vDash n} \prod_{j=1}^{\ell(\rho)} \frac{1}{(2^{s_j}-1)(r_j+1)!}$$
$$= \exp\left(-n\log n + n - \frac{\log^2 n}{2L} + C_F - \frac{\log^2 n}{2L^2 n} + \Psi(\lambda_n) + \mathcal{O}(n^{-1})\right). \tag{96}$$

The recurrence therefore has a moving periodic multiplier; no ansatz with a single fixed constant can be sharp.

9.2 The half moments

Adding Stirling's expansion to (95),

$$\log d_n = \frac{1}{2}\log(2\pi n) + C_F + \Psi(\lambda_n) - \frac{\log^2 n}{2L}$$
$$- \frac{\log^2 n}{2L^2 n} + \mathcal{O}(n^{-1}). \tag{97}$$

In particular,

$$d_n \sim \sqrt{2\pi n}\,\exp\left(C_F + \Psi(\lambda_n) - \frac{\log^2 n}{2L}\right). \tag{98}$$

The notation "$\sim$" is legitimate because the omitted logarithmic error tends to zero. The periodic factor $\exp\Psi(\lambda_n)$ remains part of the equivalent.

9.3 The literal q-binomial numerator

By Theorem 3.2,

$$\mathcal{N}_n = \mathfrak{m}_n a_n. \tag{99}$$

Furthermore,

$$\mathfrak{m}_n = 2^{\binom{n+1}{2}}(1/2;1/2)_n \tag{100}$$

and

$$\log(1/2;1/2)_n = \log(1/2;1/2)_\infty + \mathcal{O}(2^{-n}). \tag{101}$$

Consequently

$$\log\mathcal{N}_n = \frac{L}{2}n^2 - n\log n + \left(1+\frac{L}{2}\right)n - \frac{\log^2 n}{2L}$$
$$+ C_F + \log(1/2;1/2)_\infty - \frac{\log^2 n}{2L^2 n} + \Psi(\lambda_n) + \mathcal{O}(n^{-1}). \tag{102}$$

This formula is the sharp asymptotic of the finite signed q-binomial/Thue–Morse sum itself. It makes the scale cancellation visible: the numerator grows like $\exp(Ln^2/2)$, while the explicit denominator in (41) contains 2^{n^2}. The residual triangular factor is exactly the decay $2^{-\binom{n}{2}}$ in (16).

9.4 A first all-orders corollary

The general small-argument saddle expansion supplies a first correction

$$A_1(u) = \frac{1}{24} + \frac{1}{2L} - \frac{\Psi''(u) + \Psi'(u)^2}{2L^2}. \tag{103}$$

At reciprocal powers this gives

$$\log F(2^{-n}) = \mathcal{M}_n + \frac{A_1(\lambda_n)}{\lambda_n} + \mathcal{O}(\lambda_n^{-2}). \tag{104}$$

The same correction can be transported to a_n, d_n, the composition sum, and $\mathcal{N}_n$ by the exact multiplicative normalizations above. Higher orders have the same structure: a Poincaré series in inverse powers of λ_n with smooth periodic coefficient functions evaluated at the exact phase.

10 What each exact formula contributes asymptotically

It is useful to separate exact equivalence from asymptotic convenience. The formulae compute the same numbers, but they expose different structures.

Exact representation	What it exposes	Best asymptotic use
Endpoint identity $F(2^{-n}) = d_n/(n!2^{\binom{n}{2}})$	The triangular dyadic factor, factorial drift, and one unknown half moment	Immediately proves the coefficient $-L/2$ of n^2; becomes sharp after d_n is compared with $A(-n)$
Recurrence for a_n	A positive triangular coefficient recursion	Reassemble into $A(2z) = E(z)A(z)$; direct iteration term by term obscures the moving saddle
Composition sum	A positive sum over all ordered descents of the recurrence	Identifies a_n exactly; asymptotics are best obtained after resummation into A, not by selecting a few compositions
Entire product $A(z) = \prod E(z/2^j)$	The dyadic scale at every analytic level	Primary source for both the negative-Laplace harmonic sum and the coefficient saddle
Thue–Morse block	A zero of exact order k and a finite product over dyadic scales	Converts the inner signed power sum into a coefficient of $A(-2^k X)/A(-X)$
Half-base q-binomial row	A divided difference on $-1, -2, \ldots, -2^n$	Extracts the leading coefficient a_n after the block cancellation; termwise asymptotics are badly conditioned
q-Pochhammer normalization	The Mersenne product $\mathfrak{m}_n$ in normalized form	Contributes the elementary limit $(1/2; 1/2)_n \to (1/2; 1/2)_\infty$ and cancels $\mathfrak{m}_n$ exactly
Denominator and valuation formulae	Prime-adic arithmetic of the exact rationals	Strong arithmetic information, but by themselves insufficient to determine Archimedean magnitude or the periodic saddle phase

10.1 Why direct termwise analysis of the q-sum is misleading

The formula

$$F(2^{-n}) = \frac{\mathcal{N}_n}{2^{n^2}(1/2; 1/2)_n}$$

contains an explicit logarithmic contribution $-Ln^2$, whereas the final leading term is only $-Ln^2/2$. The missing $+Ln^2/2$ is generated inside the alternating finite numerator. There is no contradiction: the outer Gaussian-binomial row is designed to annihilate all lower scale powers exactly, and the surviving leading coefficient is exponentially larger than a naive bounded-summand estimate would suggest.

The cancellation can be quantified without inspecting any individual summand. From (102),

$$\log \mathcal{N}_n = \frac{L}{2}n^2 + \mathcal{O}(n \log n), \tag{105}$$

while

$$-\log\left(2^{n^2}(1/2;1/2)_n\right) = -Ln^2 - \log(1/2;1/2)_\infty + o(1). \tag{106}$$

Their sum is $-Ln^2/2 + \mathcal{O}(n \log n)$, as required.

Remark 10.1 (Numerical conditioning). A floating-point implementation of the literal signed formula is normally inferior to the positive recurrence. It must resolve the two exact annihilations before dividing by a factor of size 2^{n^2}. The decompressed identity $\mathcal{N}_n = \mathfrak{m}_n a_n$ is therefore not only a proof device; it is the stable evaluation strategy.

11 How much of the full small-argument asymptotic is encoded by inverse dyadics?

11.1 The arbitrary-argument formula

For $x \downarrow 0$, let

$$\lambda(x) = -\frac{1}{L}W_{-1}(-Lx), \qquad \lambda(x)2^{-\lambda(x)} = x. \tag{107}$$

The full endpoint saddle has the same compact main term:

$$\log F(x) = -\frac{L}{2}\lambda(x)^2 + \left(1 + \frac{L}{2}\right)\lambda(x) - \frac{1}{2}\log\lambda(x) + C_F + \Psi(\lambda(x)) + \mathcal{O}(\lambda(x)^{-1}). \tag{108}$$

At $x = 2^{-n}$ this becomes (5). Thus the exact dyadic arithmetic recovers the restriction of the entire arbitrary-argument asymptotic, including the same constant and the same periodic function at the same exact phase.

11.2 Dense sampling of the periodic correction

The sequence of inverse dyadics does not freeze the periodic phase. Put

$$y_n = \lambda_n - n = \log_2 \lambda_n. \tag{109}$$

Then y_n is increasing and unbounded, while

$$y_{n+1} - y_n \longrightarrow 0. \tag{110}$$

Indeed, $\lambda_n \sim n$ and $y_{n+1} - y_n = \log_2(\lambda_{n+1}/\lambda_n) = \mathcal{O}(n^{-1})$.

Lemma 11.1 (Density of the dyadic Lambert phases)**.** *The fractional parts* $\{\lambda_n\} = \{y_n\}$ *are dense in* $[0,1]$.

Proof. Fix $\theta \in [0,1]$. For every sufficiently large integer m, let $n(m)$ be the first index for which $y_{n(m)} \geq m + \theta$. Then

$$0 \leq y_{n(m)} - (m+\theta) < y_{n(m)} - y_{n(m)-1}.$$

The right side tends to zero by (110). Hence the fractional parts of $y_{n(m)}$ tend to θ. □

After removing the nonperiodic terms in (6), the exact rational sequence therefore determines Ψ uniquely. More precisely, if

$$R_n = \log F(2^{-n}) + \frac{L}{2}n^2 + n\log n - \left(1 + \frac{L}{2}\right)n + \frac{\log^2 n}{2L} - C_F + \frac{\log^2 n}{2L^2 n}, \tag{111}$$

then

$$R_n = \Psi(\lambda_n) + \mathcal{O}(n^{-1}). \tag{112}$$

For any $u \in [0,1]$, choose a subsequence with $\{\lambda_{n_j}\} \to u$; continuity gives $R_{n_j} \to \Psi(u)$. In this precise sense the inverse-power values encode the whole periodic correction, not just a single phase sample.

11.3 What inverse powers alone do not prove

The density result does not by itself extend (5) uniformly to all $x \downarrow 0$. Monotonicity only brackets a point $2^{-(n+1)} \le x \le 2^{-n}$ between two endpoint values whose logarithms differ by order n:

$$\log F(2^{-n}) - \log F(2^{-(n+1)}) = Ln + \mathcal{O}(\log n). \tag{113}$$

That uncertainty is enormous compared with the desired $\mathcal{O}(n^{-1})$ remainder.

The global formula (47) contains every dyadic rational and could, in principle, be used on meshes much finer than one reciprocal-power interval. To derive (108) by that route one would need a two-parameter asymptotic uniform in both numerator and denominator. The arbitrary-argument Bromwich saddle avoids that difficult finite-sum uniformity. Accordingly, the exact inverse-power formulae recover the full phase function and all constants, but a genuinely uniform analytic argument remains necessary to pass from the dense arithmetic set to a sharp theorem on a continuum.

12 Numerical verification from exact rational values

The recurrence (17) computes a_n, hence $F(2^{-n})$, in exact rational arithmetic. For example,

$$F(2^{-10}) = \frac{134926369}{1246394851358539387238350848000}, \tag{114}$$

so

$$\log F(2^{-10}) = -50.5775682823421608125\ldots.$$

Let

$$\rho_n = \log F(2^{-n}) - \mathcal{M}_n. \tag{115}$$

The leading theorem predicts $\rho_n = \mathcal{O}(1/\lambda_n)$, and the first correction predicts $\lambda_n\rho_n = A_1(\lambda_n) + \mathcal{O}(1/\lambda_n)$. The table was computed by exact rational recurrence followed by one high-precision logarithm; P and its derivatives were evaluated from the absolutely convergent product and forward-tail series.

The fourth and fifth columns approach one another, while the last column remains bounded and appears to settle, exactly as (104) predicts.

A second check displays the cancellation in the literal q-formula. Since

$$\log F(2^{-n}) = \log \mathcal{N}_n - n^2 L - \log(1/2;1/2)_n,$$

one obtains:

The Pochhammer logarithm has already converged to

$$\log(1/2;1/2)_\infty = -1.2420620948124149458\ldots. \tag{116}$$

The nontrivial compensation is between $\log \mathcal{N}_n \sim Ln^2/2$ and the explicit $-Ln^2$ denominator.

n	λ_n	ρ_n	$\lambda_n\rho_n$	$A_1(\lambda_n)$	$\lambda_n^2(\rho_n - A_1/\lambda_n)$
10	13.78503056	0.05801333439	0.7997155875	0.7629678468	0.5065687288
20	24.62186833	0.03182756966	0.7836542295	0.7629268731	0.5103462408
40	45.50804986	0.01701388954	0.7742689337	0.7629490545	0.5151456258
80	86.43351899	0.008896709914	0.7689739453	0.7629823191	0.5178773369
160	167.3870441	0.004576872612	0.7661091776	0.7630070725	0.5192522062

Table 2: Exact reciprocal-power values against the compact Lambert main and first correction.

n	$\log \mathcal{N}_n$	n^2L	$\log(1/2;1/2)_n$	$\log F(2^{-n})$
10	17.4960644003	69.3147180560	-1.2410853733	-50.5775682823
20	95.4684669707	277.2588722240	-1.2420611411	-180.5483441121
40	447.4055728988	1109.0354888959	-1.2420620948	-660.3878539023
80	1957.8744511397	4436.1419555836	-1.2420620948	-2477.0254423491

Table 3: Logarithmic cancellation in the finite q-binomial formula.

13 Correspondence with the Lean development

The relevant modules in the 25 August 2026 repository snapshot [14] form a fairly literal proof graph. The table records the role of each group rather than every import edge.

Module	Role in the bridge
`FabiusRecurrenceSequence.lean`	Defines $a_n = d_n/n!$, proves its recurrence, identifies $F(2^{-n}) = 2^{-\binom{n}{2}}a_n$, and connects its ordinary series to the analytic generating function and dyadic product.
`FabiusInverseDyadicClosedForm.lean`	Unfolds the inverse-dyadic recurrence into the exact composition formula.
`FabiusRawQBinomialFormula.lean`, `FabiusQBinomialFormula.lean`, `FabiusQBinomialScalarFormula.lean`, `FabiusDyadicQBinomialScalar.lean`	Develop the Thue–Morse block factorization, half-base Gaussian interpolation, scalar translation invariance, and the global exact dyadic formula. The coefficient-extraction argument in Section 3 is the ordinary-mathematics unpacking of this layer.
`FabiusDyadicLogBounds.lean`	Combines the exact endpoint identity with elementary half-moment bounds to prove the $-Ln^2/2$ law and an explicit $3n\log(n+1)$ error.
`NegativeLaplace.lean`, `PeriodicCorrection.lean`, `PeriodicFourier.lean`	Construct the negative-Laplace logarithm, its exact quadratic-plus-periodic decomposition, the forward-tail bound, regularity, mean normalization, and Fourier data.
`EndpointLaplaceComparison.lean`	Proves the second-order pointwise and integrated comparison between $(1-x)^n$ and $\mathrm{e}^{-nx}(1-nx^2/2)$ and transfers the estimate through the logarithm.
`DyadicSharpDecomposition.lean`, `FabiusDyadicSharpCumulant.lean`	Assemble the exact logarithmic decomposition of $F(2^{-n})$, including Stirling, the periodic term, the endpoint/Laplace error, and the exact second tilted cumulant.
`LaplacePeriodicSecondOrder.lean`	Differentiates the periodic Laplace decomposition and proves the explicit $\Psi'(\log_2 n)$ correction with an $\mathcal{O}(1/n)$ remainder.
`FabiusLambertPhase.lean`, `FabiusDyadicSharpAsymptotic.lean`	Construct the exact lower-Lambert phase, transfer the periodic derivative term to $\Psi(\lambda_n)$, and prove the corrected sharp dyadic theorem.

Module	Role in the bridge
FabiusSharpLambertMain.lean, FabiusSharpAsymptotic.lean, FabiusFullAsymptoticExpansion.lean	Prove the arbitrary-small-argument Lambert main and continue the saddle expansion to all orders.

The coefficient-saddle derivation in Section 8 is deliberately marked as an alternative exposition. The formal dyadic proof uses the endpoint–Laplace comparison; the formal arbitrary-argument proof uses the Bromwich/saddle kernel. No claim is made that the Cauchy coefficient route appears as a separate theorem under that exact name.

14 Conclusions

The conspicuous gap between exact dyadic evaluation and endpoint asymptotics can be closed without adding a new special function or an independent heuristic ansatz.

First, the exact reciprocal-power values are controlled by a single coefficient sequence:

$$F(2^{-n}) = 2^{-\binom{n}{2}} a_n, \qquad A(z) = \sum_{n \geq 0} a_n z^n = \prod_{j \geq 1} E(z/2^j).$$

The recurrence and composition formula are simply two expansions of this identity.

Second, the q-binomial formula reduces exactly to the same sequence. The inner Thue–Morse sum is a coefficient of $A(-2^k X)/A(-X)$, and the outer half-base Gaussian row is the leading-coefficient functional on the nodes -2^k. Therefore

$$\mathcal{N}_n = \mathfrak{m}_n a_n$$

with no approximation. This is the direct algebraic connection that makes asymptotic analysis of the finite q-formula feasible.

Third, the negative-axis product of A satisfies an exact dyadic dilation equation. Removing its quadratic drift leaves a smooth periodic function. A second-order comparison between the endpoint kernel $(1-X)^n$ and the Laplace kernel e^{-nX} supplies the first cumulant correction; differentiating the exact periodic decomposition identifies that correction; and the lower-Lambert fixed point moves it to the exact phase. The result is

$$\log F(2^{-n}) = -\frac{L}{2}\lambda_n^2 + \left(1 + \frac{L}{2}\right)\lambda_n - \frac{1}{2}\log \lambda_n + C_F + \Psi(\lambda_n) + \mathcal{O}(n^{-1}).$$

Finally, the inverse dyadic sequence is richer than a sparse one-phase sample: the fractional parts of λ_n are dense, so the exact rational values determine the entire continuous periodic correction after renormalization. What they do not provide by monotonicity alone is a uniform continuum theorem at the sharp error scale; that step still requires a uniform saddle argument or a two-parameter analysis of the general dyadic formula.

The answer to the motivating question is therefore affirmative in a strong sense. The small-argument asymptotic can be re-derived from the exact calculation formulae, including the constant, the logarithmic correction, the exact lower-Lambert phase, and the nonconstant periodic fluctuation. The only essential move is to respect the exact cancellations built into those formulae rather than attempting to estimate their finite summands independently.

A Algebra behind the expanded dyadic formula

This appendix records the expansion needed to pass from the compact Lambert form to the explicit expression in n. Put $\ell = \log n$ and seek

$$\lambda_n = n + \frac{\ell}{L} + \frac{c_1}{n} + \frac{c_2}{n^2} + \mathcal{O}(\ell^3/n^3). \tag{117}$$

The fixed-point equation is

$$\lambda_n = n + \frac{1}{L}\log\lambda_n. \tag{118}$$

Writing

$$\lambda_n = n\left(1 + \frac{\ell}{Ln} + \frac{c_1}{n^2} + \frac{c_2}{n^3} + \cdots\right)$$

and expanding the logarithm gives

$$\log\lambda_n = \ell + \frac{\ell}{Ln} + \frac{c_1 - \ell^2/(2L^2)}{n^2} + \mathcal{O}(\ell^3/n^3).$$

Comparison in (118) yields

$$c_1 = \frac{\ell}{L^2}, \qquad c_2 = \frac{\ell - \ell^2/2}{L^3}, \tag{119}$$

which gives the first four terms of (79).

Now set

$$H(\lambda) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log\lambda.$$

Substitution of (117) with (119) and collection through order $1/n$ gives

$$\begin{aligned} H(\lambda_n) = &-\frac{L}{2}n^2 - n\ell + \left(1+\frac{L}{2}\right)n - \frac{\ell^2}{2L} \\ &- \frac{\ell^2}{2L^2 n} + \mathcal{O}(n^{-1}). \end{aligned} \tag{120}$$

The notation $\mathcal{O}(n^{-1})$ in (120) includes bounded multiples of $1/n$; the displayed ℓ^2/n term is separated because it is asymptotically larger. Adding $C_F + \Psi(\lambda_n)$ proves (86).

B A stable numerical recipe

The following procedure reproduces Table 2 without evaluating the unstable signed q-sum.

Algorithm B.1 (Exact arithmetic plus convergent products)**.**

1. Compute $a_0 = 1$ and, for $1 \le j \le n$,

   $$a_j = \frac{1}{2^j - 1}\sum_{k<j}\frac{a_k}{(j-k+1)!}$$

   in rational arithmetic.

2. Form the exact rational

   $$F(2^{-n}) = 2^{-\binom{n}{2}}a_n$$

   and take its logarithm only after conversion to the desired floating-point precision.

3. Solve $\lambda - \log\lambda/L = n$ on the large branch, either by $\lambda = -W_{-1}(-L2^{-n})/L$ or by Newton iteration initialized with $n + \log n/L$.

4. Reduce $u = \lambda - \lfloor \lambda \rfloor$ and put $s = 2^u \in [1, 2)$. Evaluate

$$P(u) = \sum_{j \geq 1} \log \frac{1 - \mathrm{e}^{-s/2^j}}{s/2^j} + \frac{L}{2}(u^2 - u) + \sum_{m \geq 0} \log(1 - \mathrm{e}^{-s2^m}).$$

   The first series has a geometric small-argument tail and the second a doubly exponential tail. Use stable primitives such as `expm1` for the first factors.

5. Subtract $\overline{P}$ from (61) to obtain $\Psi(u)$. Differentiate the locally uniformly convergent series termwise, or use the rapidly convergent Fourier series (63), to obtain Ψ' and Ψ''.

6. Evaluate $\mathcal{M}_n$ from (85) and, when desired, A_1 from (103).

For a direct verification of the q-formula one may compute $\mathcal{N}_n = \mathfrak{m}_n a_n$ exactly and compare it with the finite signed expression. The equality is a better test than comparing floating-point approximations of the two sides because it checks both cancellation layers over the rationals.

C Notation cross-reference

Symbol	Meaning
L	$\log 2$
F	bounded Fabius function / distribution function on $[0, 1]$
up	Rvachev up-function, with $\mathrm{up}(t) = F(1 - \lvert t \rvert)$ on $[-1, 1]$
X	random variable with distribution function F
d_n	half moment $\mathbb{E}(1 - X)^n$
a_n	factorially normalized half moment $d_n/n!$
$A(z)$	$\sum a_n z^n = \mathbb{E}\mathrm{e}^{zX}$
$E(z)$	$(\mathrm{e}^z - 1)/z$
$\mathfrak{m}_n$	$\prod_{j=1}^{n}(2^j - 1)$
$\mathcal{N}_n$	finite q-binomial/Thue–Morse numerator for $F(2^{-n})$
$B(s)$	$A(-s) = \mathbb{E}\mathrm{e}^{-sX}$
$\Lambda(s)$	$\log B(s)$
P	exact one-periodic correction in Λ
Ψ	$P - \overline{P}$, mean-zero periodic correction
C_F	constant $\overline{P} - \frac{1}{2}\log(2\pi)$
λ_n	large solution of $\lambda - \log \lambda / L = n$
$\mathcal{M}_n$	compact Lambert main term for $\log F(2^{-n})$

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